

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

CHE 222 — Past Questions & Materials
by Renance

Course: CHE 222
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 2
Levels: 200

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Paper 1 of 2 — CHE 222 Chemistry For engineers

Level 200 • Second Semester • PastQuestion

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COLLEGE OF ENGINEERING & TECHNOLOGY

DEPARTMENT OF-CHEMICAL ENGINEERING

28? SEMESTER EXAMINATION, 2023/2024 ACADEMIC SESSION

COURSE TITLE: CHEMISTRY FOR ENGINEERS

COURSE CODE: CHE 222 COURSE UNIT: 2 UNIT TIME: 2 HOURS

INSTRUCTION: There are six (6) Questions in all. Answer only four questions. All questions carry equal marks. Start each question on a fresh page

(a) (1) State three (3) areas of application of Chemistry in Chemical or Petrochemical Engineering.

Gi) List two (2) Chemical industries that apply the concepts in Chemistry

(b) Using the Band theory of solids, classify solid materials into Insulator, Semiconductor, and Conductors.

(c) A chemical plant wants to produce 500 kg of sodium dodecylbenzenesulfonate (SDBS), a key component in detergent manufacturing. Starting with benzene as a raw material. design a sulfonation process and calculate the required amount of sulfuric acid to achieve 20% conversion. :

2.

(a) Define the following

(1) Open System (ii) Closed System (iii) Isolated system (iv) Isentropic Process

(v) Isothermal Process

(b) (i) Using two typical examples, explain the statement “the Perpetual motion machine of the first order is impossible”

(ii) Classify the following properties as either INTENSIVE OR EXTENSIVE properties: Mass. heat capacity, temperature, chemical potential, specific heat

(c) One (1) mole of liquid benzene is completely

burnt in oxygen to form liquid water and carbon dioxide gas, the enthalpy

of reaction $\Delta H = -78 \text{ kcal}$ at 25°C (298K). Calculate the heat of reaction at constant volume at the same temperature

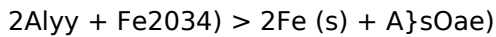
* $2 \text{ cal/ deg C} \cdot \text{mol}$]

GW) Define the following terms relating to thermochemistry.

(i) Heat of Solution (ii) Heat of f

ormation (iii) Heat of Neutralization (iv) He
reaction (v) Bond energy
at of

(b) Calculate the standard heat of reduction of Ferric oxide by Al
uminum which proceeds
according to the following reaction:



Additional information

Heat of formation of Fe_2O_3 and Al_2O_3 are -196.5 kcal/mole and -399.1 kcal/mole respectively.

(c) In a typical Fluid catalytic cracking unit
(FCCU), the catalyst activity is reduced, leading to low
product yield and selectivity. List four possible causes of this
problem

4.

(a) (i) Define a Spontaneous Process (ii) Describe the spontaneity using the Gibbs's free energy

(b) (i) Define the law of mass action
spontaneity or otherwise of a process

(i) A typical reaction in gaseous State is shown below:
 $a\text{A} + b\text{B} \rightleftharpoons c\text{C} + d\text{D}$

P_A , P_B , and P_C are the partial pressures of various gaseous species at equilibrium,
show that $K_p = K_c(RT)^{\Delta n}$.

Hint: K_p is the equilibrium constant in terms of partial pressure

K_c is the equilibrium constant in terms of concentration

(c) Water dissociates according to the following reaction:



At 1500°C , the value of K_p

is 1.87×10^{-5} , Assuming ideal behavior

the corresponding value of K_c (Equilibrium concentration in mole/liter)

Hint: $R = 8.314 \text{ J/K mol}$

For a mixture of gases, calculate

K_c (constant in terms of concentration) in

0.08205 L.atm/degC mole

(i) State the Le-Chatelier's Principle

: define the following: (1) Homogeneous (2) Heterogeneous

(iii) Extent of reaction

(iv) Limiting reactant (v) Excess reactant

(vi) Activation Energy (vii) Equilibrium constant

(viii) Define (1) Limiting reactant

(2) Activation Energy (3) Equilibrium constant

(c) The following data was obtained for a reaction:

(T CO)

(i) 4 3

(ii) $k \times 10$ (sec)

LL

(iv) Oxidation (v) Isomerization

& blocks of the Petrochemical Industry

(c) (i) Identify three (3) areas for the

give

(ii) A Chemical Plant is experiencing issues with foam formation during Production

Process leading to reduced Product yield. Suggest a Suitable Surfactant

how it can be used to control foam,

Paper 2 of 2 — CHE 222 (chemistry for engineers)

Level 200 • Second Semester • PastQuestion

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COLLEGE OF ENGINEERING AND TECHNOLOGY

DEPARTMENT OF CHEMICAL ENGINEERING

Second Semester Examination, 2022/2023 Academic Session

COURSE TITLE: Chemistry for Engineers

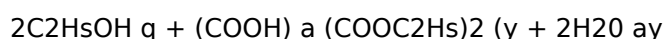
COURSE CODE: CHE 222 Credit Units: 2

INSTRUCTIONS: Answer Any Four Questions Time Allowed: 2 Hours

1, (a) i) Distinguish between heterogeneous and homogeneous Chemical Equilibria

ii) State the criteria that a reaction must satisfy in a calorimeter study.

(b) 200 ml of 0.6 molar ethanol and 300 ml of 1.5 molar (COOH)₂ were reacted together according to the equation below;



Calculate the equilibrium constant, if the equilibrium conversion is 70 %.

2. a) i) Define the Le — Chatelier's principle in a chemical equilibrium system

ii) List the usefulness of Le — Chatelier principle in a chemical equilibrium system

(b) The degree of dissociation of 4.80 moles of hydrogen iodide at 448 °C was found to be 22 %. Calculate:

(i) number of moles of hydrogen (ii) number of moles of hydrogen at equilibrium (iii)

calculate the equilibrium constant for the dissociation reaction. The reaction is given as;

$2HI \rightleftharpoons H_2 + I_2$

3. (i) Describe the substitution reactions of benzene with electrophiles

(ii) Illustrate with chemical reaction the Friedel — Crafts alkylation of benzene

(iii) With the aid of chemical reactions describe the mechanism of Friedel — Crafts alkylation with primary haloalkanes.

(iv) Define aromatic sulfonation of benzene

4., (a) Define colloids and briefly explain how they differ from solutions. Provide colloidal systems.

(b) Briefly explain the role of adsorption in catalysis and why it is essential in this context:

(c) Why is selectivity important in catalytic reactions in petrochemical refining?

(d) A chemical plant is experiencing issues with foam formation during the production process, leading to reduced product yield. Suggest a suitable surfactant and explain how it can be used to control foam.

5. (a) Define adsorption in the context of surface phenomena and provide an example of its practical significance.

(b) A cosmetic company wants to develop a new sunscreen product that offers both water and

oil resistance. Suggest a suitable emulsifier and explain how it can be used to stabilize the emulsion.

(c) Briefly explain the significance of hydrocarbons and functional groups in organic chemistry and their relevance to chemical engineering processes.

6. (a) Describe the role of catalysis in petrochemical refining and provide an example of a catalytic process used in this industry.

(b) Define wetting in the context of surface phenomena and provide an example of its industrial importance.

(c) Highlight two advantages of using emulsions in the food industry.

(d) Explain the difference between a sol and an emulsion in colloidal systems. Provide examples of each and discuss their industrial significance.